3.9 A 1.2V High-Voltage-Tolerant Bootstrapped Analog Sampler in 12-bit SAR ADC Using 3nm GAA's 0.7V Thin-Gate-Oxide Transistor

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Recently, as process refinement continues, the latest 3nm Gate-All-Around (GAA) process utilizing the Multi-Bridge-Channel-FET (MBCFET) with nanosheet technology enhances performance, power efficiency, and area (PPA) benefits by large effective channel width and enhanced design flexibility compared to the previous FinFET technology [1]. However, the 3nm GAA process provides only thin-gate oxide transistors for a 0.7V low supply voltage and does not offer thick-gate oxide transistors for higher supply voltages. Typically, a general-purpose analog-to-digital converter (ADC) needs to sample inputs at higher levels, such as 1.2V, over the 0.7V low supply voltage (VDD) in the 3nm GAA process. However, a conventional bootstrapped sampler can sample inputs only up to VDD level, using 2×VDD internal node voltages. In the previous FinFET processes with thick-gate oxide transistors for a 1.2V supply voltage, conventional bootstrapped samplers are useful enough to sample 1.2V input without any reliability and leakage issues. However, since the 3nm GAA process provides only thin-gate oxide transistors for the 0.7V low supply voltage, this paper introduces an innovative tolerant bootstrapped sampler architecture capable of sampling input levels up to 1.2V (1.7×VDD) with maximum internal node voltages of 1.9V (2.7×VDD) without reliability and leakage issues. Moreover, this paper also proposes a 0.7V 12-bit SAR ADC in a 3nm GAA process, using this new tolerant bootstrapped sampler.

Figure 3.9.1 shows the conceptual scheme and circuit implementation of the conventional bootstrapped sampler [2], which is useful to sample analog inputs equal to or lower than the supply voltage level. Generally, since the V_{GS} and V_{GD} voltages of transistors should not exceed the supply voltage (VDD) for reliability, the conventional bootstrap sampler to sample VDD level input has additional M6 and M9 transistors to ensure that the V_{GD} voltage of M4 and M5 does not exceed VDD. However, if the conventional bootstrap sampler in the 3nm GAA process with only thin-gate oxide transistors tries to sample large inputs exceeding VDD, reliability and leakage problems occur because the V_{GD} and V_{GS} of some transistors exceed VDD. As shown in Fig. 3.9.1, in the pre-charge (hold) phase, if the V_{IN} voltage is 1.2V, the gate voltages of MS and M3 are 0V. Therefore, MS and M3 have V_{GD} values exceeding 1.2V, which is greater than VDD, resulting in reliability issues. In the sampling phase, if the V_{IN} voltage is 1.2V, the voltage at the bottom plate of the C_B (node B) becomes 1.2V (1.7×VDD) and the voltage at the top plate of the C_B (node A) becomes 1.9V (2.7×VDD). Therefore, V_{GD} voltages of the M1 and M2 also exceed VDD. Additionally, since the V_{GS} voltage of M4 becomes VDD, gate voltage of the MS and M3 also becomes 2.7×VDD through M4 and the V_{GD} voltage of the M6 also exceeds VDD. Furthermore, if the V_{IN} (1.2V) voltage is higher than VDD, the drain-body diode of the M7 transistor becomes forward-biased. Consequently, leakage current occurs through V_{IN} through M3, M9, and M7. Therefore, in the 3nm GAA process, the conventional bootstrap sampler cannot sample large inputs exceeding VDD without reliability and leakage issues.

Figure 3.9.2 and Fig. 3.9.3 show a proposed tolerant bootstrapped sampler architecture to sample large analog input over the 0.7V supply voltage, utilizing only thin-gate oxide transistors. As shown in Fig. 3.9.2, in the pre-charge phase, the top plate of the C_B capacitor (node A) and gate of the dual-protection switches (M10, M11, M14, and M15) become VDD. This ensures that even when the V_{IN} voltage is 1.2V (1.7×VDD) over VDD, the V_{GD} voltage of M3 and MS does not exceed VDD. As shown in Fig. 3.9.3, in the sampling phase, if the V_{IN} voltage is 1.2V, the voltage at the bottom plate of the C_B (node B) becomes 1.2V (1.7×VDD) and the voltage at the top plate of the C_B (node A) becomes 1.9V (2.7×VDD). To prevent the V_{GD} voltage of M1 from exceeding VDD, the gate of M14 should be connected to VDD. Similarly, to ensure that the V_{GD} voltage of M2 does not exceed VDD, the gate of M15 should be connected to the bottom plate (node B) of the C_B. Adding a C1 capacitor to node C and node B ensures that when V_{IN} is lower than VDD, the node C maintains VDD. Conversely, when V_{IN} exceeds VDD, the node C approximately follows the V_{IN} voltage. Therefore the node P is approximately V_{IN} and the V_{GD} of M6 does not exceed VDD, because the node C is the gate of M16. The proposed bootstrap sampler removes M7 in Fig. 3.9.1 to block the leakage current when the V_{IN} voltage exceeds VDD. Instead of M7, as shown in Fig. 3.9.3, it employs a new dynamic floating level shifter to control the gate voltage of M4. Compared to the conventional floating-level-shifter structure with the fixed level-shifting voltage [4], the new dynamic floating level shifter changes the level-shifting voltage according to the V_{IN} voltage level. The dynamic floating level shifter connects the sources of M22 and M23 to the top plate of the C_B capacitor and connects the gate of M24 to the bottom plate of the C_B capacitor. In the sampling phase, the gate voltage of the M25 is V_{IN} voltage and the V_{GD} voltages of M24 and M26 do not exceed VDD, because the node D and node C follow V_{IN} voltage. Additionally, the proposed bootstrapped sampler prevents temporal reliability issues by adding M17-M21 transistors and C_2 capacitors. During the transition time between the sampling and the pre-charge phases, the node B, O, and F can be discharged to 0V, while the node E is connected to the node A (VDD+V_{IN}). If the M17-M21 transistors do not exist, V_{GD} or V_{GS} voltage of M4, M8, M9, M24, and M25 can become higher than VDD with reliability risk. However, the M17-M21 transistors ensure that the transistors connected to the node B, E, and O do not exceed VDD even during the temporarily transition time. The C2 capacitor also protects the M24 and M25 transistors by preventing the rapid discharge node F.

Figure 3.9.4 shows the 0.7V 12-bit SAR ADC in the 3nm GAA process. It consists of the proposed bootstrapped input samplers, reference sampling switches, a capacitor DAC array, a pre-amplifier, a comparator, and logic blocks. To sample and convert a 1.2V input in the ADC, the capacitor DAC requires two-stacked single-protection reference sampling switches, since the capacitor DAC array needs to sample either 0V or 1.2V reference voltage. According to the comparison results between the ADC input and reference voltages, the SAR logic sequentially generates D_n (n=1,2,…,12) signals. In the reference sampling switches, the M1 transistor samples reference bottom voltage (0V) using the D_n signal or the M3 transistor samples reference top voltage (1.2V) according to the $D_{n,up}$ signal, which is a level-up signal of D_n in the 0.5V~to~1.2V range, generated by the limited level shifter. M2 with a VDD gate voltage and M4 with a V_{MID} (0.5V) gate voltage are protection transistors when the sampled output ($Vout_n$) voltages are at 1.2V and 0V levels, respectively. For stable 12-bit ADC performance, the ADC incorporates an additional pre-amplifier. Even though the pre-amplifier consumes static current, the signal gain and offset sampling scheme of the pre-amplifier can remove a complex offset calibration scheme in the comparator. Moreover, a signal attenuation capacitor (C_{ATT}) makes the pre-amplifier inputs less than VDD for reliability.

The 3nm GAA process ADC using the proposed sampler occupies 0.027mm² as shown in Fig. 3.9.7. The power consumption of the ADC is 0.34mW at 2MS/s based on 0.7V supply and 1.2V reference top voltages. Figure 3.9.5 shows the measured differential nonlinearity (DNL), integral nonlinearity (INL) and FFT spectrum of the ADC with a 1.2V peak-to-peak and 0.1MHz analog input. The peak DNL and INL are 0.47LSB and 0.6LSB, respectively. At 2MS/s, the measured SFDR and SNDR are 84dB and 63dB, respectively. Figure 3.9.6 compares the proposed bootstrapped sampler with the previous ones [2,3,5]. The conventional bootstrapped sampler with a voltage doubler [2] and the alternating bootstrapped sampler without a voltage doubler [5] can only sample up to the input signals at the same level as the supply voltage. The structure presented in [3] can sample input signals over the supply voltage but relies on drain-extended MOS (DEMOS) to endure high-voltage stress. The proposed tolerant bootstrapped sampler can sample up to 1.2V level input over the 0.7V supply voltage by using only 3nm GAA process thin-gate oxide transistors without reliability and leakage issues.

References:

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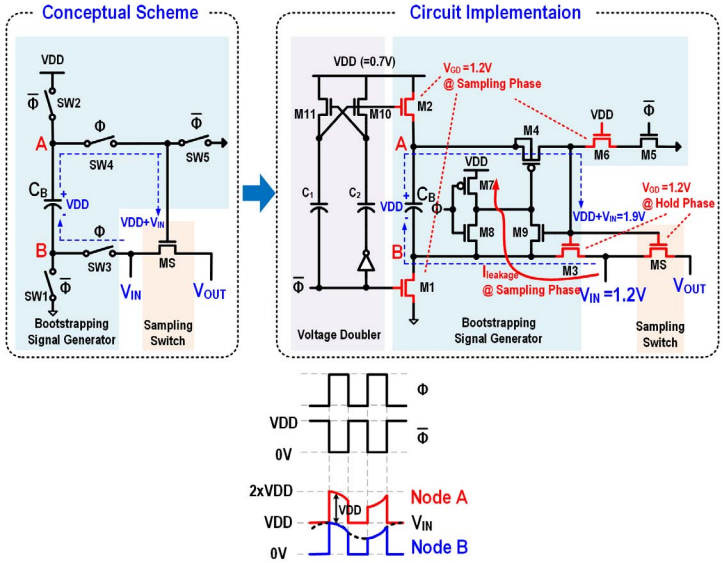


Figure 3.9.1: Conventional bootstrapped sampler architecture.

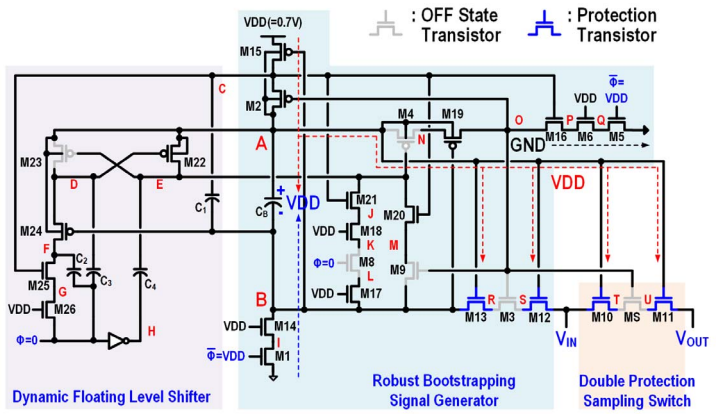


Figure 3.9.2: Pre-charge (hold) phase operations of the proposed tolerant bootstrapped sampler.

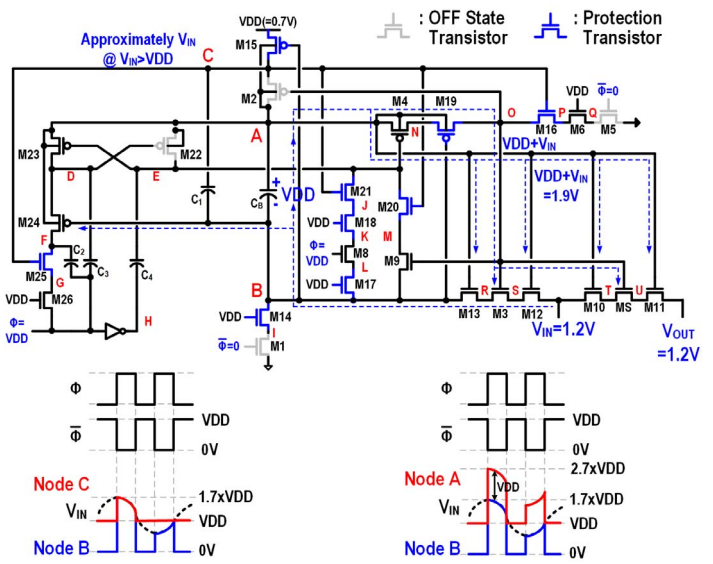


Figure 3.9.3: Sampling phase operations of the proposed tolerant bootstrapped sampler.

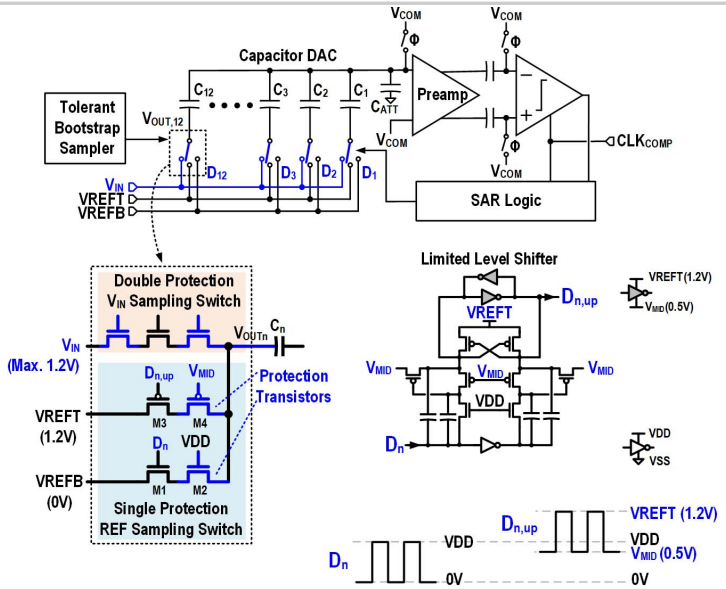


Figure 3.9.4: Protection sampling switches and limited level shifter in SAR ADC.

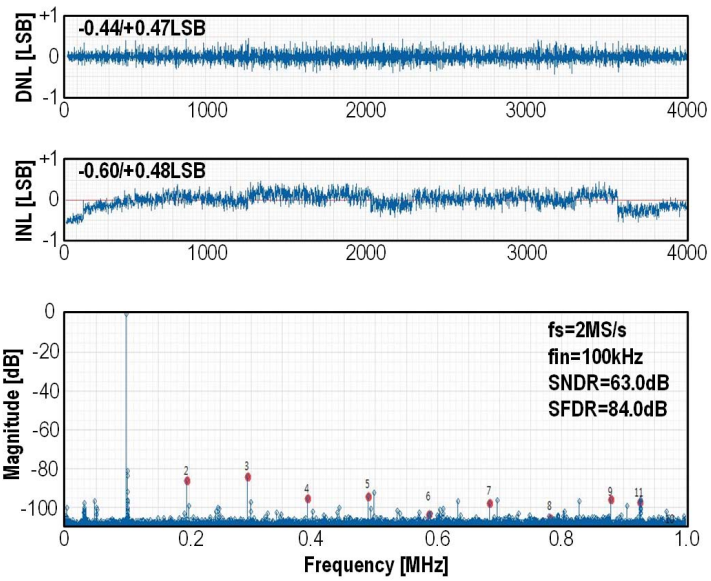


Figure 3.9.5: Measured DNL, INL and FFT spectrum.

	This work	[2] JSSC 1999	[3] JSSC 2006	[5] JSSC 2010
Feature of Bootstrapped Sampler	**Over the Supply Voltage w/ Tolerant Design**	Conventional w/ Voltage Doubler	Over the Supply Voltage w/ DEMOS	Conventional w/o Voltage Doubler
Technology	**3nm**	600nm	350nm	130nm
Supply Voltage [V]	**0.7**	1.5	2.75	1.2
Input swing [V_{pp}]	**1.2**	1.5	5.5	1.2
Transistor	**Thin-gate oxide**	Thick-gate oxide	Thick-gate oxide and DEMOS	Thick-gate oxide

Figure 3.9.6: Comparison with the previous bootstrapped samplers.

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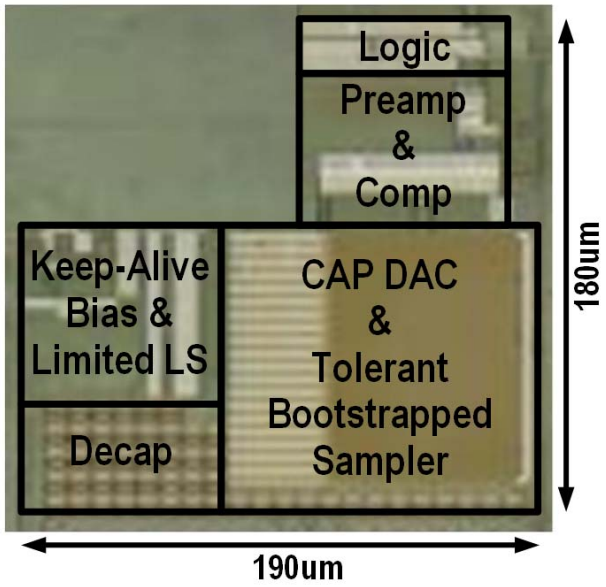


Figure 3.9.7: Die micrograph.

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